



Semester Two Examination, 2019

Question/Answer booklet

MATHEMATICS APPLICATIONS UNITS 1 AND 2

Section One: Calculator-free

If required by your examination administrator, please
place your student identification label in this box

Student number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Section One: Calculator-free**35% (52 Marks)**

This section has **eight (8)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1**(6 marks)**

(a) Simplify $2 + 4 \times \sqrt{36} - 20 \div 2^2$.

(2 marks)

(b) Determine the value of the expression $\frac{x^2 + 2x}{y^2 - 3y}$ when $x = 5$ and $y = -2$.

(2 marks)

(c) Determine the value of d when $a = 2.8$, $v = 0.5$ and $d = 6v^2 - 2av$.

(2 marks)

Question 2

(6 marks)

An airport car park charges the amounts shown in the table below for periods not exceeding 24 hours.

Parking time	No more than 3 hours	More than 3 but not exceeding 6 hours	Each additional hour (or part) exceeding 6	Maximum charge for up to 24 hours
Charge	\$25	\$35	\$5	\$65

(a)

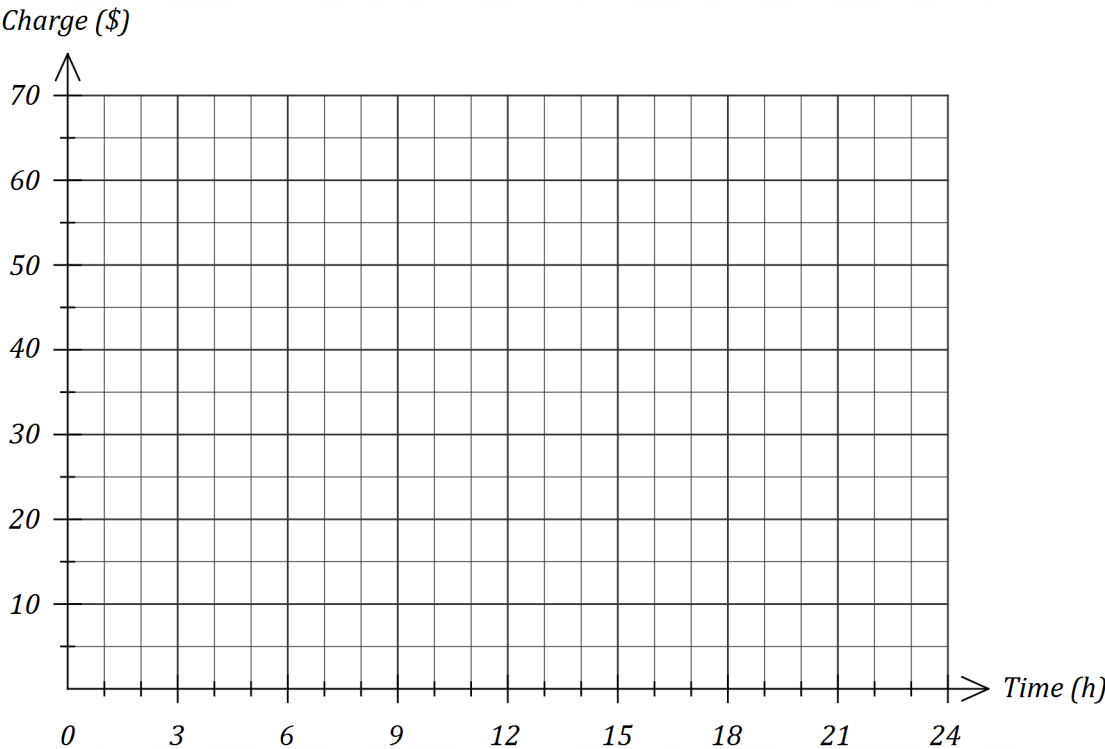
Explain why the charge to park a car for eight and a half hours is \$50.

(2 marks)

(b)

Draw a graph to represent the parking charges on the axes below.

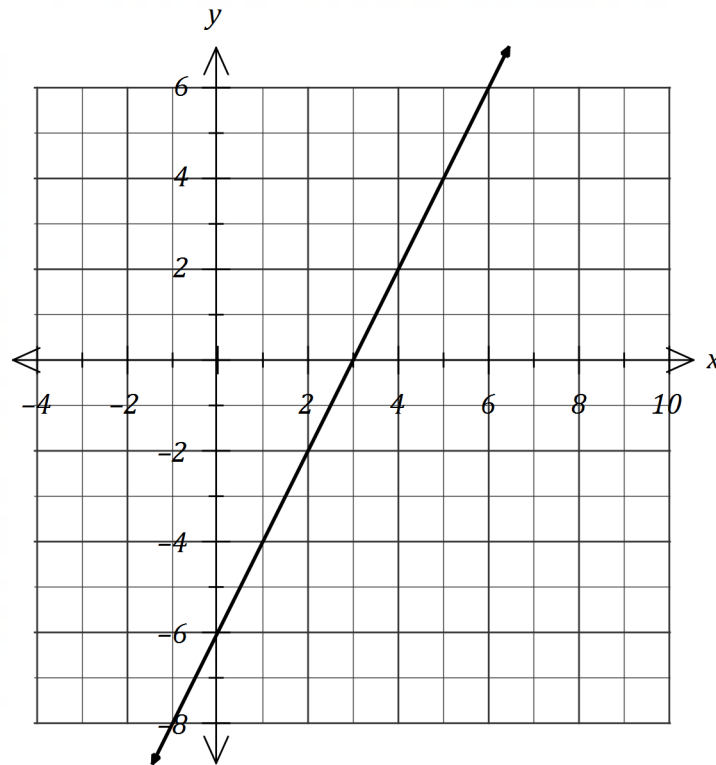
(4 marks)



Question 3

(7 marks)

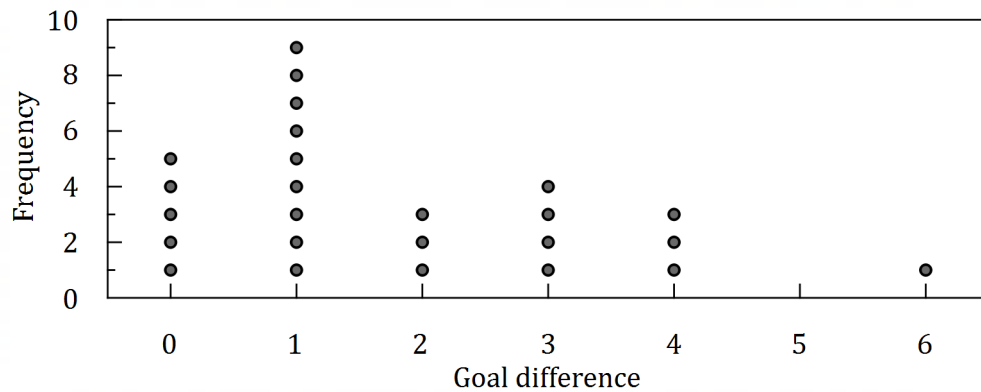
The graph of line L_1 with equation $y = ax + b$ is shown below.



- (a) Determine the value of the constant a and the value of the constant b . (2 marks)
- (b) Straight-line L_2 has equation $x + 2y = 8$.
- (i) Draw L_2 on the axes above. (2 marks)
- (ii) State the gradient of L_2 . (1 mark)
- (c) Determine the equation of the straight-line L_3 that passes through the origin and the point of intersection of lines L_1 and L_2 . (2 marks)

Question 4**(9 marks)**

The dot plot below shows the goal difference between each pair of teams playing in a weekend round of a soccer league.



(a) State the number of games that did not end in a draw. (1 mark)

(b) Classify the type of variable that goal difference is by circling two of the following words: (1 mark)

Categorical. Continuous. Discrete. Nominal. Numerical. Ordinal.

(c) Determine

(i) the median goal difference. (1 mark)

(ii) the mean goal difference. (2 marks)

(d) Describe the distribution of the dataset in terms of modality, shape and location. (3 marks)

(e) Describe the feature of the dot plot that would suggest the median would be less than the mean of the dataset. (1 mark)

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Question 5**(7 marks)**

The number of minutes that a group of students took to complete a task are listed below:

29, 26, 17, 29, 18, 26, 31, 25, 26, 35, 33, 28

- (a) Construct an ordered stem-and-leaf plot to display this data. **(3 marks)**

- (b) Identify, with justification, any outliers that may be present in the times. **(4 marks)**

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Question 6**(5 marks)**

(a) Solve the equation $3(x + 5) - 1 = x - 2(x - 3)$.

(2 marks)

(b) Use the formula $A = \frac{1}{2}(a + b)h$ to determine

(i) A when $a = 5.6$, $b = 8.4$ and $h = 20$.

(1 mark)

(ii) b when $A = 33$, $a = 4$ and $h = 6$.

(2 marks)

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Question 7

(6 marks)

(a) Simplify $2 \begin{bmatrix} 5 & 2 \\ -3 & 4 \end{bmatrix} - \begin{bmatrix} 6 & -1 \\ 2 & 7 \end{bmatrix}$.

(2 marks)

(b) Given that $A = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$ determine A^2 .

(2 marks)

(c) Determine the value of x and the value of y given that $\begin{bmatrix} 2 \\ 5 \end{bmatrix} + x \begin{bmatrix} -3 \\ 4 \end{bmatrix} = \begin{bmatrix} -10 \\ 3y \end{bmatrix}$.

(2 marks)

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Question 8**(6 marks)**

At the start of the year 2002, the difference in the ages of a mother and her child was 24.

Let the age of the mother be x and the age of her child be y at the start of 2002.

- (a) Use the above information to write an equation relating x and y . (1 mark)

Three years later, the mother was 5 times as old as her child.

- (b) Use this additional information to write another equation relating x and y . (2 marks)

- (c) Determine the age of the child at the start of the year 2019. (3 marks)

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Semester Two Examination, 2019

Question/Answer booklet

**MATHEMATICS
APPLICATIONS
UNITS 1 AND 2**

Section Two:

Calculator-assumed

If required by your examination administrator, please
place your student identification label in this box

Your teacher's name _____

Your name _____

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper,
and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Section Two: Calculator-assumed

65% (98 Marks)

This section has **thirteen (13)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9

(6 marks)

A person who moved to Australia exchanged 80 000 euros for Australian dollars and placed the proceeds into a 3-month term deposit.

At the time of the exchange, 1 euro bought 1.6067 Australian dollars.

- (a) Determine the amount the person deposited in the term deposit. (1 mark)

- (b) The term deposit paid simple interest of 1.5% per annum. Calculate the interest earned in the deposit over the 3 months. (2 marks)

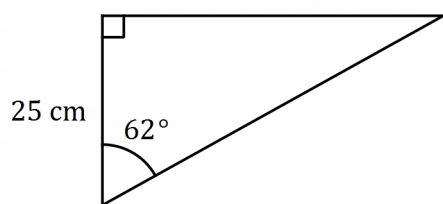
- (c) After another currency exchange, the person placed \$75 000 into a savings account paying 2.6% p.a. interest compounded monthly. Determine the interest that accumulated in this account during the first 6 months. (3 marks)

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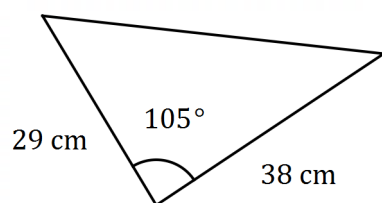
Question 10**(6 marks)**

Determine, with justification, the area of each of the following triangles.

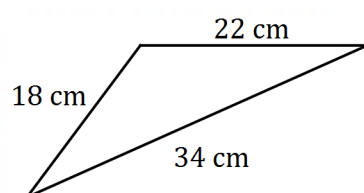
(a)

**(2 marks)**

(b)

**(2 marks)**

(c)

**(2 marks)**

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Question 11

(7 marks)

A survey of the ages, x , of 300 football fans at a recent match gave rise to the following data.

Age group	Frequency
$18 \leq x < 30$	54
$30 \leq x < 40$	60
$40 \leq x < 50$	63
$50 \leq x < 60$	75
$60 \leq x < 80$	48

- (a) Explain why it is necessary to use the interval midpoint of each age group when calculating the mean age of the fans. (1 mark)
- (b) State the interval midpoint for the age group $60 \leq x < 80$. (1 mark)
- (c) Determine the mean and standard deviation of the ages of the fans in the survey, rounding your values to 2 decimal places. (3 marks)

It was later discovered that of the 48 fans aged 60 or more, one was aged 75 and all the others were younger than 64.

- (d) Use this information to determine better estimates for the mean and standard deviation of the ages of the fans in the survey. (2 marks)

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Question 12**(11 marks)**

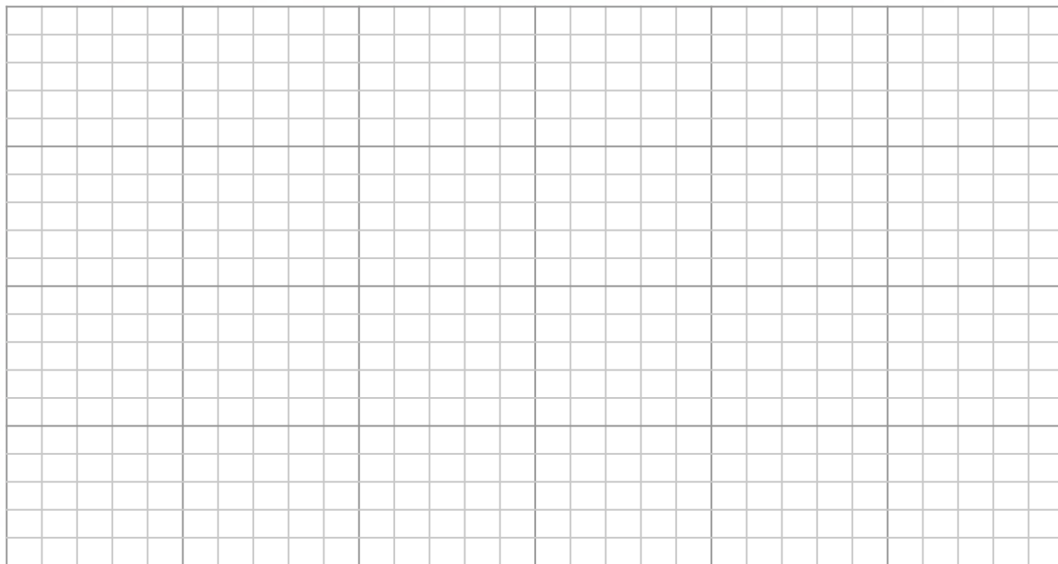
Every month a small business runs 15 information seminars for prospective customers. The number of people attending the seminars in two consecutive months is summarised below, where the largest turnout for a June seminar was 82 people.

Attendances in May (LHS) and June (RHS)

7 4	5	
7 3 2	6	2 4 6 9 9
9 9 7 0	7	1 2 3 4 4 6 7 8
9 5	8	1 2
9 9 7 4	9	

(a) Determine the range of attendance figures for May. (2 marks)

(b) Construct parallel boxplots for the May and the June attendances on the grid below. (6 marks)

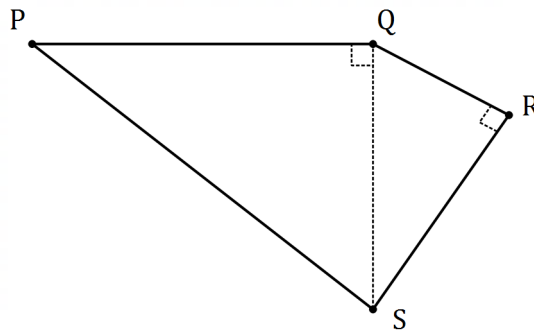


- (c) Compare the attendance figures for May with those of June. (3 marks)

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Question 13**(7 marks)**

A field is bounded by four straight fences PQ , QR , RS and SP as shown below, where $QR = 120$ m, $RS = 160$ m, $SP = 290$ m and $\angle PQS = \angle QRS = 90^\circ$.



- (a) Determine the distance QS . (1 mark)
- (b) The fence around the field needs maintenance at a cost of \$1.15 per metre. Determine the cost of this maintenance. (3 marks)
- (c) The field was recently sprayed with a treatment at a total cost of \$382.50. Calculate the cost, in cents per square metre, of this treatment. (3 marks)

Question 14

(8 marks)

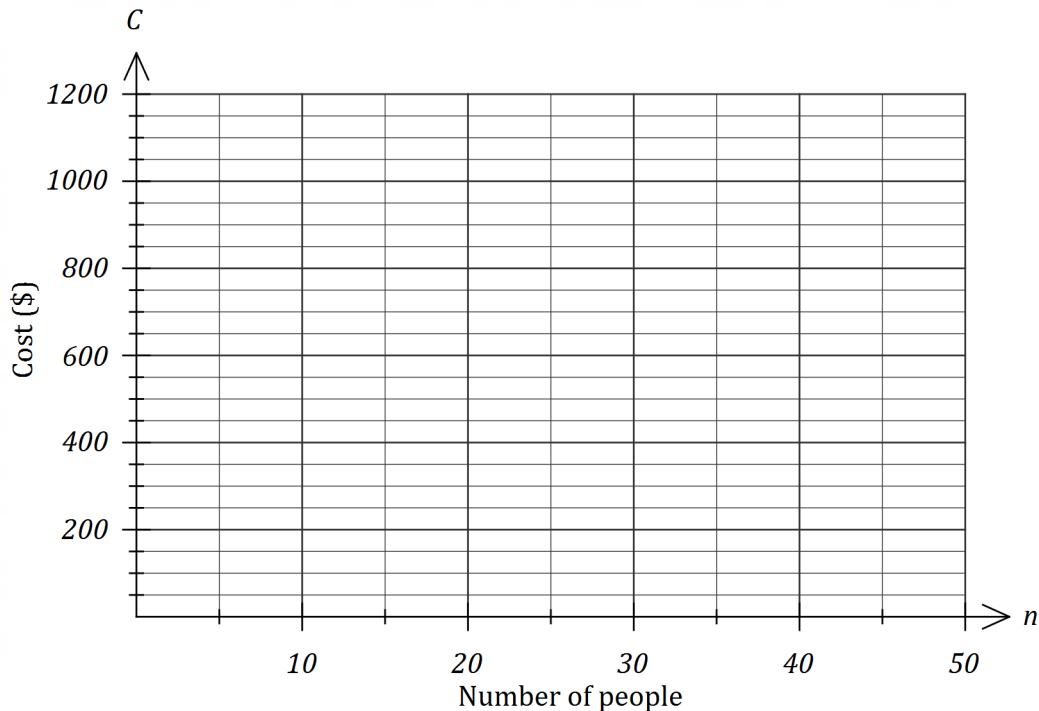
A person is considering a choice of two caterers to supply snacks for an office party. Caterer *A* charges \$20 per person plus a once-off booking fee of \$150.

(a) Calculate the cost of using caterer *A* when

(i) 10 people are expected to attend. (1 mark)

(ii) 40 people are expected to attend. (1 mark)

(b) Plot the two catering costs from (a) on the axes below and draw a straight line through them. (2 marks)



Caterer *B* simply charges \$25 per person.

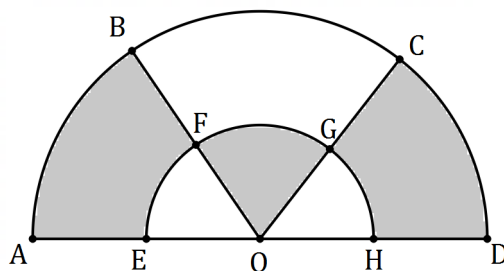
(c) Add a line to the graph above to represent the cost of using caterer *B*. (2 marks)

(d) Write a brief statement to the person recommending which caterer to use if minimising the cost was the only consideration. (2 marks)

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Question 15**(8 marks)**

The shaded areas in the diagram below form a logo for a business. $OABCD$ is a semicircle with radius $OA = 30$ cm and $\angle AOB = \angle BOC = \angle COD$. The inner semicircle has radius $OE = 14$ cm.



- (a) Determine the area of the sector OFG . (2 marks)
- (b) Determine the total shaded area. (3 marks)
- (c) Determine the shaded area in a copy of the logo that is enlarged by a scale factor of 5 for use on a flag, giving your answer in square metres. (3 marks)

Question 16**(7 marks)**

The lifetimes of a type of heating element are normally distributed with a mean of 455 hours and a standard deviation of 32 hours.

- (a) State the percentage of elements that are expected to have a lifetime within two standard deviations of the mean. (1 mark)
- (b) Determine the standard score for an element that lasts for 433 hours. (2 marks)
- (c) Determine the probability that a randomly selected element will have a lifetime
- (i) of less than 433 hours. (1 mark)
- (ii) of more than 510 hours. (1 mark)
- (iii) within 25 hours of the mean lifetime. (2 marks)

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Question 17**(8 marks)**

Three small huts lie on level ground. Hut B lies 450 m due north of hut A and hut C lies 570 m from hut A on a bearing of 125° .

(a) Sketch a diagram to show this information. (2 marks)

(b) The huts are equipped with radios that have a range of 800 m. Showing use of trigonometry, determine whether huts B and C can communicate by radio. (3 marks)

(c) Showing use of trigonometry, determine the bearing of hut C from hut B . (3 marks)

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Question 18

(9 marks)

A shop usually sells the same brand of AAA batteries in packs of 4, 8 and 14 for \$7.95, \$13.90 and \$19.92 respectively, but currently has the 4-packs and 8-packs on sale at 20% off.

- (a) Determine the total price and hence the average price per battery for 18 batteries when a customer buys a 4-pack and a 14-pack in the sale. **(3 marks)**

- (b) Use the sale prices to rank the pack sizes in order of value from best to worst. **(3 marks)**

- (c) The shop buys a carton containing 36 of the 14-packs from a wholesaler for \$420.48 excluding GST. The shop then adds their profit margin of $k\%$ and another 10% GST to arrive at the advertised pack price above. Determine the value of k . **(3 marks)**

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Question 19**(6 marks)**

Entry fees at a mini-golf course for children, adults and seniors are \$6.50, \$12.00 and \$7.25 respectively. The following table shows the breakdown of the number of paying customers at the course over three days.

Day	Children	Adults	Seniors
Friday	12	63	51
Saturday	35	82	27
Sunday	42	75	33

- (a) Use two matrices to write a calculation that will result in matrix M , where M shows the total entry fees collected on each of the three days and determine M . (3 marks)
- (b) Matrix M can be multiplied by matrix N to produce a 1×1 matrix that shows the sum of all entry fees collected over the three days. Write down N . (1 mark)
- (c) Use two matrices to write a calculation that will result in matrix P , where P shows the total number of patrons in each fee category over the three days and determine P . (2 marks)

Question 20**(7 marks)**

Employees at a card yard can opt to have their gross wage calculated in one of two ways:

- Option A - commission of 1.6% of their weekly sales.
- Option B - commission of 1.1% of their weekly sales plus \$350 per week.

(a) Determine the gross weekly wage for an employee choosing

(i) option A in a week when their sales were \$125 500. (1 mark)

(ii) option B in a week when their sales were \$78 250. (1 mark)

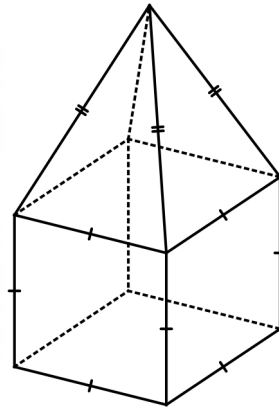
(b) The gross weekly wage for an employee using option B was \$812. Determine the weekly sales this employee made. (2 marks)

(c) Explain which option is best, depending on an employee's weekly sales. (3 marks)

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Question 21**(8 marks)**

A solid monument built on level ground has the form of a pyramid mounted on a cubical base with sides of length 4.8 m as shown below (not to scale). The vertex of the pyramid is 4.5 m directly above the centre of its square base.



- (a) Determine the volume of the monument, to the nearest cubic metre. **(3 marks)**

- (b) Determine the total surface area of the monument, excluding its square base that rests on the ground. **(5 marks)**